Boing Network: Design Plan

> 👋 **Everyday users:** why Boing is its own L1 — not a fork. Short version: SIX-PILLARS.md.
> 🛠️ **Developers:** architecture, tokenomics, and design decisions. Spec-level truth is TECHNICAL-SPECIFICATION.md.
> 🛰️ **Operators:** stack choices (Rust, libp2p, HotStuff) constrain how you run nodes.

Core Principle: An independent L1 built from first principles — optimized for efficiency, free from dependencies on other chains, designed for **absolute decentralization**, and committed to **transparency by default** in how we build, govern, and operate.
Quick overview: BOING-NETWORK-ESSENTIALS.md.

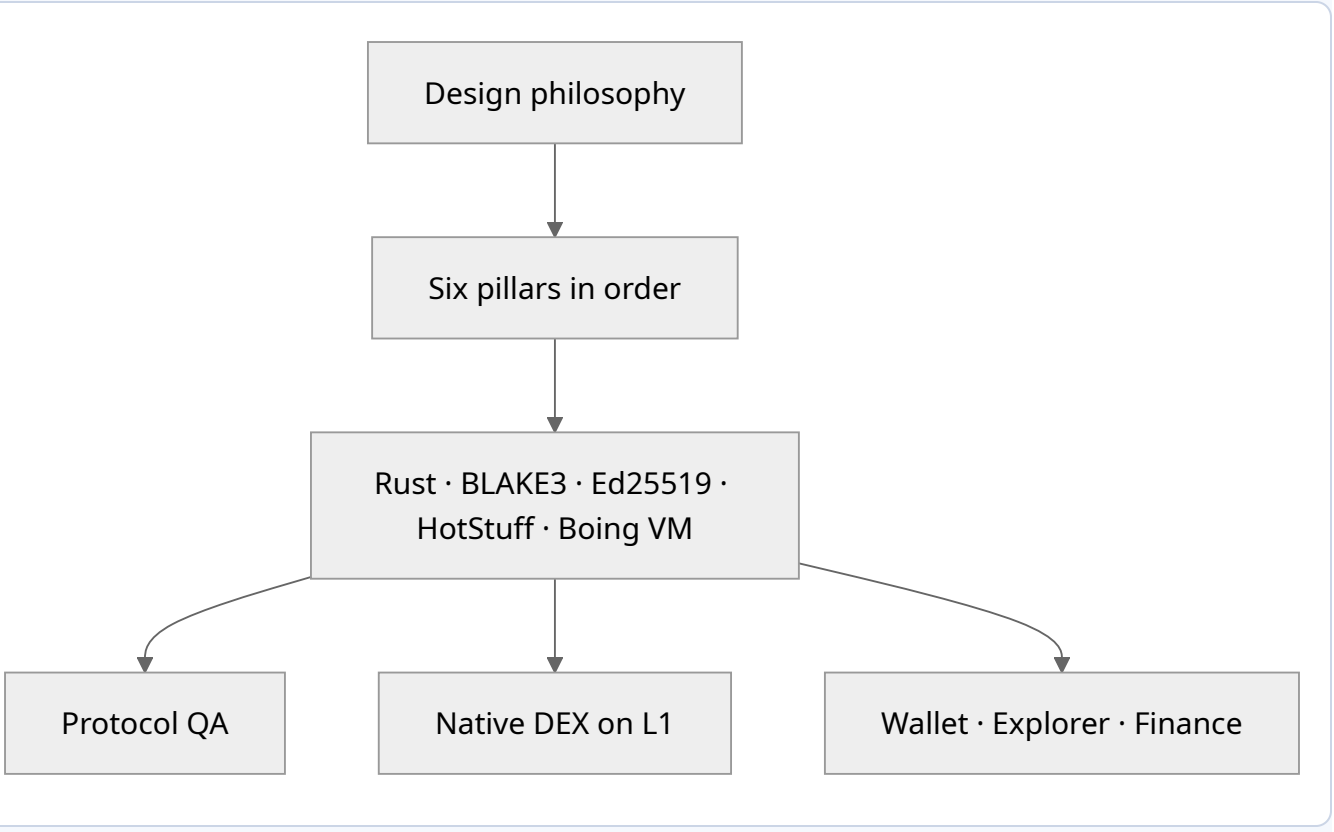


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Design Philosophy

The Boing blockchain is conceived as a **greenfield L1** with the following pillars:

- **Authentic** — Your own architecture, not a fork or framework
- **Independent** — Core protocol does not depend on another L1 for consensus, execution, or identity
- **Optimal** — Adopt the best ideas from the ecosystem when they're demonstrably superior
- **Unique** — A distinct identity and technical story
- **Decentralized** — Absolute decentralization as a foundational requirement
- **Transparent** — Transparent by default in how we build, govern, and operate — trust through verifiability, not promises

Transparency: 100% Open by Design

Principle: Community trust is earned through transparency. We commit to 100% transparency in everything we do and how the network is built. No hidden parameters, no black boxes, no “trust us.”

Commitment	Implementation
Protocol design	Open design docs, public specs, auditable code. Anyone can verify how the network works.
Tokenomics	Emission schedules, fee splits, burn mechanics — all on-chain and documented. No surprise minting or parameter changes without governance.
Governance	On-chain proposals, votes, and outcomes. No backroom deals; all decisions visible and auditable.
Validator & staking	Clear slashing conditions, reward formulas, and distribution. Delegators see exactly how rewards flow.
Treasury & spending	Public accounts, transparent allocation, governance-approved disbursements. Community can trace every BOING.
Security & audits	Audit reports public. Known risks disclosed. No security-through-obscurity.
Development	Open source. Roadmap, progress, and trade-offs discussed openly. Community can contribute and verify.

Why it matters: Trust cannot be decreed — it must be earned. Transparency allows the community to verify that the network behaves as stated, that no single party can act without accountability, and that the rules are enforced fairly. 100% transparency is the foundation for a community that can trust.

Priority Pillars

Boing Network prioritizes, in order:

1. **Security** — Safety and correctness over speed
2. **Scalability** — Throughput and efficient resource use
3. **Decentralization** — Permissionless participation at every layer
4. **Authenticity** — Unique architecture and identity
5. **Transparency** — 100% openness in design, governance, and operations — the foundation for community trust
6. **True quality assurance** — Top-notch standards with built-in automation: only assets that meet protocol-defined rules and security bar are allowed on-chain. All **currently known** edge cases are resolved by the automated regulatory QA system (Allow or Reject); **leniency for meme culture** (meme/community/entertainment are valid purposes), with **no maliciousness or malignancy** allowed (blocklist and scam checks enforced). The community QA pool is only for genuinely unknown or policy-mandated cases. See QUALITY-ASSURANCE-NETWORK.md.

Innovation: Unique Boing Features

Boing aims to introduce **distinctive functionalities** that differentiate it from other blockchains. The following are candidate innovations. See also UX & Human-Centered Innovation and Technical Innovations for Ecosystem Improvement.

1. Native Account Abstraction (Built-in)

Innovation	Description	Differentiation
Native AA	Account abstraction built into the protocol (not a contract layer). Users can have programmable accounts (gasless, social recovery, session keys) without additional infrastructure.	Many stacks implement AA in contracts or auxiliary mempools rather than the base protocol. Boing would make AA a first-class protocol primitive.

2. Adaptive Gas Model

Innovation	Description	Differentiation
Dynamic gas	Gas costs adapt to network congestion and resource usage in real time. Predictable upper bounds with fair prioritization.	Most chains use static gas tables. Boing could introduce a more responsive, fair pricing model.

3. Native Cross-Chain Primitives

Innovation	Description	Differentiation
Protocol-level bridges	Light client verification and cross-chain messaging as protocol primitives, not third-party contracts.	Bridges today are typically trust-minimized contracts. Boing could design cross-chain into the base layer.

4. Protocol-Enforced Quality Assurance (No Other Chain Does This)

Innovation	Description	Differentiation
Built-in QA automation	Only deployments that meet defined standards (bytecode rules, security heuristics) are accepted. Reject at submit/mempool and execution; no "deploy anything" by default.	Most contract L1s allow any bytecode; standards are conventions only. Boing enforces at the protocol.
Community QA pool	When automation is unsure, a pool of community members votes Allow/Reject. Transparent, time-bounded, with clear outcomes.	Edge cases are handled without a single gatekeeper; aligns with decentralization.
Single source of truth	A clear set of rules and attributes each asset type must satisfy; updatable via governance.	Developers and users know exactly what is allowed on-chain.

See QUALITY-ASSURANCE-NETWORK.md for the full design.

5. Delegated Staking with Slashing Transparency

Innovation	Description	Differentiation
Transparent slashing	Clear, auditable slashing conditions with appeal mechanisms. Delegators see exactly why validators were slashed.	Improves trust and decentralization of staking.

6. On-Chain Governance with Time Locks

Innovation	Description	Differentiation
Phased governance	Parameter changes go through time-locked stages (proposal → cooling → execution) to allow for community response and exit.	Reduces risk of sudden, controversial upgrades.

7. Verifiable Delay + Ordering (Boing-specific VDF)

Innovation	Description	Differentiation
Custom VDF	A verifiable delay function for fair ordering that doesn't depend on any external chain's PoH. Boing-native, audit-friendly design.	Unique to Boing; enables fair ordering without external dependencies.

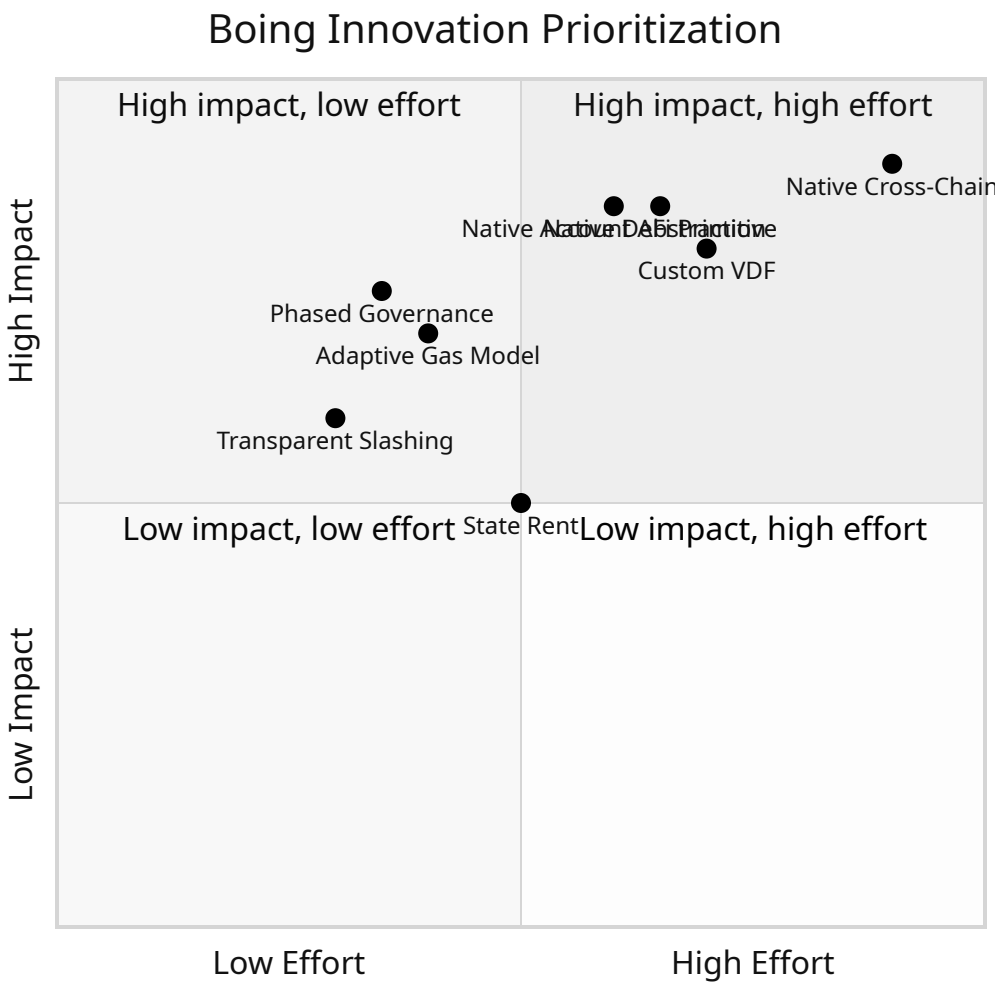
8. State Rent with Grace Periods

Innovation	Description	Differentiation
Rent with grace	Inactive accounts pay state rent, but with long grace periods and clear notifications before eviction. Balances state bloat with user experience.	More user-friendly than abrupt state eviction.

9. Native DeFi Primitives (boing.finance Integration)

Innovation	Description	Differentiation
Protocol-level AMM	A minimal, audited AMM or swap primitive built into the protocol for boing.finance to use. Reduces contract risk.	Most DeFi is fully contract-based. Boing could offer a secure, protocol-level primitive.

Innovation Prioritization Matrix



Unique Use-Case: Cross-Chain DeFi Coordination Layer

Boing's distinct role in the blockchain ecosystem is to serve as a **DeFi coordination layer** that reduces fragmentation, improves composability, and benefits all chains — not just Boing itself.

The Problem Boing Addresses

- **Fragmented liquidity** — Value is spread across dozens of chains; users and protocols must bridge, wrap, and duplicate.
- **Trust-heavy bridges** — Most cross-chain infrastructure relies on multisigs or opaque relayers.
- **Poor DeFi UX** — Users juggle multiple wallets, RPCs, and chains to access liquidity.
- **Isolated innovation** — New chains launch in silos; interoperability is an afterthought.

Boing's Unique Position

Boing is the native chain for cross-chain DeFi coordination — a hub where:

Capability	How It Benefits the Ecosystem
Protocol-level cross-chain primitives	Light client verification, not third-party relays. Any chain can connect trust-minimally.
Unified liquidity routing	boing.finance as the coordination front-end; Boing Network as the settlement and routing layer. Reduces fragmentation for users across all chains.
Native DeFi primitives	Protocol-level AMM, token deployment, and swap logic. Audited, upgradeable, and composable by design.
Account abstraction at base layer	Gasless UX, social recovery, session keys — built in. Other chains bolt these on; Boing ships them.

How This Serves the Broader Ecosystem

- **For users:** One place to deploy tokens, trade, and route liquidity across chains with minimal bridging friction.
- **For developers:** A chain with native cross-chain hooks; build once, compose everywhere.
- **For other chains:** A coordination partner, not a competitor. Boing aggregates liquidity and routes it back; other chains gain access to a larger pool.
- **For DeFi protocols:** A settlement layer for cross-chain swaps, lending, and composability that doesn't depend on a single bridge operator.

Sustainability Through Use-Case

The use-case drives sustainability: if Boing reduces friction and increases composability, usage grows. Usage drives fees and demand for BOING. Fees fund validators and the treasury. The system funds itself through **real utility**, not speculation or unsustainable incentives.

UX & Human-Centered Innovation

Principle: Blockchains are powerful technically, but UX is stuck in the early-internet era. Boing aims to make blockchain feel like a modern product, not a cryptography lab. UX is the multiplier that unlocks adoption, security, and developer velocity.

Frictionless UX + Invisible Infrastructure

Current Friction	Boing Direction
Managing private keys	Native Account Abstraction; embedded wallets; social recovery
Understanding gas fees	Gasless transactions (paymasters); predictable pricing
Switching networks	Automatic chain routing; users don't need to know which chain they're on
Handling wallets, seed phrases, RPC endpoints	Unified identity; human-readable flows; minimal configuration
Dealing with failed transactions	Simulation + pre-confirmation; predictable outcomes
Inconsistent UI patterns	SDK and standards for consistent onboarding

Trust & Verification ("What am I signing?")

Problem	Innovation
Users can't understand transactions	**Human-readable intent** — "Send 100 BOING to Alice" instead of hex blobs
Contract legitimacy unknown	**Simulation + attestation** — Pre-flight simulation; on-chain or off-chain reputation
Phishing and impersonation	**Verified displays** — Clear address resolution; contract metadata

Recoverability (Beyond Seed Phrases)

Problem	Innovation
Lost seed phrase = permanent loss	**Social recovery** — Guardians, time locks; protocol-level, not bolt-on
No familiar account recovery	**Institutional custody options** — MPC, threshold custody; opt-in for users who want it

Cost Predictability & MEV

Problem	Innovation
Gas spikes; failed txs still cost gas	**Adaptive gas model** — Predictable upper bounds; simulation before submit
Frontrunning, sandwich attacks	**MEV-aware design** — Private mempool options; encrypted transactions; fair ordering (VDF)

Developer Experience (DX)

Problem	Innovation
Fragmented tooling	**Unified SDK** — One SDK for contracts, clients, and cross-chain
Different chain APIs	**Consistent RPC** — Open standards; same patterns across Boing and bridges
Testing across chains	**Local/devnet parity** — Deterministic testnets; cross-chain simulation

Regulatory & Compliance UX

Problem	Innovation
Enterprises need compliance	**Privacy-preserving attestations** — ZK proofs for KYC/AML; selective disclosure
Compliance destroys pseudonymity	**Compliant rails** — On-ramps and attestations that don't require full identity on-chain

Off-Ramps & Real-World Integration

Problem	Innovation
Getting value out is hard	**Fiat off-ramps** — Partner integrations; stablecoin redemption paths
Payments to merchants	**Payment rails** — Protocol-level primitives for real-world payments

How This Amplifies Everything Else

- **Scalability** — Meaningful only when people can use the apps
- **Security** — Improves when users aren't forced into risky key-management patterns
- **Developer velocity** — Increases when onboarding friction disappears
- **Enterprise adoption** — Faster when workflows feel familiar

Technical Innovations for Ecosystem Improvement

Beyond UX, Boing can contribute **technical innovations** that benefit the broader blockchain ecosystem:

1. Light Clients & Stateless Verification

Innovation	Benefit
Verkle-based stateless clients	Lighter nodes; more participants; better decentralization
Compact proofs	Smaller bandwidth; faster sync; mobile-friendly validation

2. Trust-Minimized Cross-Chain Primitives

Innovation	Benefit
Light client verification at protocol level	Bridges without multisig relayers; any chain can verify Boing (and vice versa)
IBC-style or custom interoperability	Open protocol for cross-chain messaging; ecosystem-wide composability

3. MEV Mitigation & Fair Ordering

Innovation	Benefit
VDF-based ordering	Fair, verifiable ordering; reduces frontrunning and sandwich attacks
Encrypted mempool options	User-controlled; transactions revealed only at execution
Commit-reveal schemes	For sensitive operations; MEV-resistant by design

4. Intent-Based Architecture

Innovation	Benefit
Intent signing	Users sign high-level goals ("swap X for Y at best price"); solvers execute
Human-readable transactions	Easier verification; fewer phishing and approval mistakes

5. Transaction Simulation & Pre-Confirmation APIs

Innovation	Benefit
Pre-flight simulation	Clients get outcome before submit; no surprise failures
Pre-confirmation commitments	"This tx will succeed or you don't pay" — improves UX across ecosystem

6. Open Standards & Interoperability

Innovation	Benefit
Open RPC / API standards	Consistent interfaces; easier integration for wallets, explorers, dApps
Shared bridge standards	Other chains can adopt; reduces fragmentation
Modular execution/DA	Optional separation of data availability and execution; flexibility for rollups

7. Cryptographic & Protocol-Level Improvements

Innovation	Benefit
Efficient ZK proofs	For bridges, privacy, and compliance; smaller proofs, faster verification
BLS aggregation	Compact multi-signatures; less bandwidth for consensus
Threshold cryptography	Social recovery; distributed key management

Ecosystem Impact

These technical innovations are designed to be **exportable** — specifications, reference implementations, and protocols that other chains and projects can adopt. Boing improves the ecosystem not only by being a great chain, but by contributing reusable building blocks.

Success-Based dApp Incentives, Boing SDK & Decentralized Automation

Boing's developer ecosystem is built on three pillars: **success-based dApp incentives**, a **seamless Boing SDK**, and **native decentralized automation** for everyone.

Success-Based dApp Incentives

Goal: Incentivize developers to build dApps and environments *on* Boing — rewards are automated, decentralized, and based on **success** (usage, not just deployment).

Success Metrics (On-Chain, Verifiable)

Metric	What It Measures
Transaction count	Active usage
Volume (e.g. USD)	Economic activity
Fees generated	Value contributed to the network
Unique users	Reach
TVL / liquidity	Depth and stickiness

Incentive Model (Automated & Decentralized)

- **Formula:** `dapp_reward = f(metrics)` — e.g. proportional to fees generated by the dApp's contracts
- **Execution:** Smart contract computes rewards from on-chain data; distributes on schedule (e.g. monthly)
- **No gatekeepers:** Formula is governance-approved once; no per-project human approval
- **Funding:** Developer Treasury (e.g. X% of fees or block rewards) allocated by formula
- **Value cap (per dApp owner):** Each dApp owner has a **maximum incentive cap** per epoch (e.g. per month). Prevents concentration; ensures fair distribution across many builders.

Value Cap (Per dApp Owner)

Parameter	Purpose
Per-dApp cap	Maximum BOING (or equivalent) a single dApp owner can receive per epoch
Governance-controlled	Cap can be adjusted via phased governance
Transparent	Cap is on-chain; everyone knows the rules
Above cap → overflow	Amount above cap flows back to treasury or is redistributed to other dApps

Example: Cap = 10,000 BOING/month per dApp. If your dApp earns 15,000 by formula, you receive 10,000; 5,000 goes to treasury or other developers. Ensures no single dApp captures all incentive flow.

Sustainability

- Cap prevents runaway incentive cost
- Formula + cap are programmable; governance can tune over time
- Rewards align with real usage; speculation is limited

Boing SDK — Seamless Developer Experience

Goal: One SDK that makes building on Boing as smooth as modern web development.

Capabilities

Component	Purpose
`boing init`	Scaffold project with config, contracts, tests
`boing dev`	Local chain + funded wallet + hot reload; one command
`boing deploy`	Deploy to testnet/mainnet; environment-aware
`boing test`	Run tests against local or testnet
Unified client	RPC, transactions, queries — single API
Type-safe bindings	Generated from contracts; autocomplete, validation
Metrics registration	Register contracts for success-based rewards
Cross-chain helpers	Optional bridges; consistent API
Automation SDK	Schedule tasks; register triggers; conditional logic

Developer Journey

```
boing init my-dapp
cd my-dapp
boing dev          # Local chain + faucet
# Build, test
boing deploy       # Testnet or mainnet
boing metrics register --contract 0x...  # Opt into success incentives
```

Decentralized Automation as a Network Objective

Goal: Boing is a network where **anyone** — developers, validators, average users — can use decentralized automation without a central operator.

Automation Types

Actor	Use Cases
Developers	Scheduled deploys, CI/CD triggers, monitoring, auto-responses
Validators	Auto-compound, auto-rebalance, automated governance participation
Average users	DCA, recurring payments, "when X happens do Y" (e.g. buy when price drops)
dApps	Auto-rebalancing, liquidation checks, keeper tasks

Technical Components

Component	Purpose
Native scheduler	Protocol-level cron; scheduled execution
Trigger-based execution	"When event X, run Y" — on-chain or cross-chain
Conditional logic	If/else based on state, price, time
Decentralized executors	Anyone can run automation; rewarded for correct execution
User-facing automation	No-code / low-code flows (Zap-style); users set rules, network executes

Relation to Existing Solutions

Today: Chainlink Automation, Gelato, Keeper Network — third-party services. Boing aims for **protocol-native** automation: scheduling, triggers, and executor incentives built into the chain. Reduces dependency on external operators; improves decentralization and UX.

How It All Fits Together



Bootstrap & Validator Incentives (Sustainability-First)

Principle: Bootstrap incentives must not compromise long-term health. No massive early inflation, no unsustainable airdrops, no validator rewards that collapse after a cliff.

Sustainability Rules (Non-Negotiable)

Rule	Rationale
Inflation must trend toward zero (or a low long-term floor)	Prevents token dilution and preserves value for early participants.
Fee revenue must cover validator costs at maturity	The chain pays for itself through usage, not emissions.
No reward cliffs	Smooth emission curves; no sudden drops that cause validator exit.
Treasury funds ecosystem, not ongoing infra	Treasury for grants, audits, tooling — not perpetual validator subsidies.

Tokenomics: Sustainable Design

```
Emission curve:
  Year 1:  X% annual inflation (bootstrap validators)
  Year 2:  X × 0.7
  Year 3:  X × 0.5
  ...
  Year 10+: 1–2% or floor (fee revenue dominant)
```

- **Block rewards** — Decline over time; fee share increases as a proportion of validator income.
- **Fee split** — 70–80% to validators, 20–30% to treasury (grants, security, ecosystem).
- **Validator commission** — 5–10% on staking rewards; validators earn more by attracting delegators and providing quality infra.

Bootstrap Incentives That Don't Sacrifice Sustainability

Mechanism	How It Works	Why It's Sustainable
Testnet incentive program (planned; not currently running)	Possible future design: tokens or NFTs for validators, bug hunters, early builders. Non-dilutive (testnet tokens) or small mainnet allocation.	Does not inflate mainnet supply. Builds community and tests infra.
Fair launch / no pre-mine	All BOING enters via block rewards and fees. No large pre-allocations.	Fair distribution; long-term alignment.
Validator grants (one-time)	Ecosystem fund grants to early validators for hardware or hosting — capped, one-time.	Helps bootstrap; not perpetual subsidy.
Fee buyback / burn	Optional: portion of fees used to buy and burn BOING. Reduces supply as usage grows.	Usage → deflationary pressure → value accrual.

Validator Economics at Maturity

At steady state, validator revenue should come primarily from:

1. **Transaction fees** — From swaps, deployments, and cross-chain activity.
2. **Block rewards** — Low, stable inflation (e.g. 1–2%) as a base.
3. **Commission** — From delegators.

If fee revenue consistently exceeds validator costs, the network is **self-sustaining**. Inflation becomes secondary.

Phased Incentive Strategy

Phase	Validator Support	Sustainability Check
Testnet (incentives planned for the future; none currently running)	Possible testnet tokens, small grants for early validators	No mainnet dilution
Mainnet launch	Higher initial block rewards (declining curve)	Emissions programmed to decrease
Growth	Fee revenue grows; block rewards decrease	Fee share > emission share
Maturity	Fees dominant; low inflation floor	Network pays for itself

Running on a Single Machine (Low-Cost Bootstrap)

- Use your extra computer as a validator during testnet and early mainnet.
- Power cost: ~$5–20/month.
- As the chain gains usage, validator rewards (fees + block rewards) can cover this.
- No need for cloud spend until you choose to scale.

BOING Tokenomics — Sustainable Value Design

Working drafts (Proposed vs Undecided; no TGE date): BOING-INITIAL-TOKENOMICS.md · plain language: BOING-TOKENOMICS-AND-INCENTIVIZED-TESTNET.md.

Objective: Ensure BOING token value is sustainable long-term. No dilution traps, no reward cliffs, value accrual through usage and scarcity.

Supply & Emission

Parameter	Value	Rationale
Supply model	**Uncapped, floor-triggered waves**	Start with specific initial supply; when circulating supply reaches a defined floor, a new wave of tokens is minted to restore balance (see Chosen Model: Uncapped Supply with Floor-Triggered Waves).
Initial supply	Specific amount at mainnet launch (TBD)	No infinite mint at genesis; predictable starting point.
Block time	~2 seconds (target when transactions are pending; blocks are produced only when transactions are present)	Balance between throughput and finality

Emission Schedule (Block Rewards)

Epoch / Year	Annual Inflation	Notes
Year 1	8%	Bootstrap validators
Year 2	6%	Decay factor 0.75
Year 3	4.5%	
Year 4	3.4%	
Year 5	2.5%	
Year 6–10	2% → 1.5% (linear decay)	
Year 10+	**1% floor** (or 0% if fees sufficient)	Fee revenue dominant; emissions optional

Formula: `emission_year_n = emission_year_1 × (decay)^(n−1)` with `decay ≈ 0.85`

Fee Economics

Allocation	Share	Purpose
Validators	80%	Primary revenue; covers infra
Treasury	20%	Grants, audits, ecosystem
Burn	0% (optional)	In the floor-triggered wave model, burn is optional; see Burn in a Floor-Triggered Model. If enabled, governance-bounded (e.g. 0–5%).

Fee flow: Fees go to validators and treasury. With uncapped supply and floor-triggered waves, value accrual comes from usage and sustainable rebalancing rather than from burn.

Staking & Validator Economics

Parameter	Target	Rationale
Min stake	Accessible (e.g. 10,000 BOING)	Broad validator participation
Staking yield	5–15% early, 3–8% at maturity	Competitive but not Ponzi
Commission range	5–10% typical	Validator revenue; delegators keep most
Unbonding period	14–21 days	Security; prevents instant exit

dApp Incentive Caps (Value Protection)

Parameter	Value	Rationale
Per-dApp cap per epoch	Governance-controlled (e.g. 50,000 BOING/month)	Prevents single dApp capturing all incentives
Total dApp pool	% of fees (e.g. 5% of treasury share)	Bounded; doesn't dilute stakers
Overflow	To treasury or burn	Excess flows back; no waste

Deflationary Mechanisms (Optional in Floor-Triggered Model)

1. **Fee burn** — Optional (0% in base design). If enabled, governance-bounded. In a floor-triggered model, burn mainly accelerates how often waves are minted; see Burn in a Floor-Triggered Model.
2. **Floor-triggered waves** — When supply reaches the floor, new waves are minted to rebalance; no permanent deflation required.

3. **dApp cap overflow** — Unclaimed incentives → treasury (or optional small burn if governance enables it).

Value Accrual Loops

```
Usage ↑ → Fees ↑ → Validator revenue ↑ → Security ↑ → Trust ↑ → Usage ↑
Usage ↑ → Fees ↑ → Treasury ↑ → Ecosystem ↑ → Usage ↑
(Optional burn, if enabled: Usage ↑ → Fees ↑ → Burn ↑ → supply pressure; in floor-triggered model, waves then i
```

Sustainability Guardrails (On-Chain)

Guardrail	Implementation
Uncapped supply, floor-triggered waves	No hard cap; waves mint when supply reaches floor; rules on-chain and transparent
Fee split	Governance can adjust within bounds (e.g. validators 70–90%, treasury 10–30%; burn 0% or optional 0–5%)
dApp cap	Governance parameter; transparent, auditable
No reward cliffs	Wave size and trigger rules predictable; no sudden dilution

Summary: Why BOING Value Is Sustainable

1. **Uncapped supply with floor-triggered waves** — Predictable rebalancing; no arbitrary infinite inflation.
2. **Fee-dominant revenue** — Validators and treasury earn from usage; waves only when supply reaches the floor.
3. **Burn optional** — No required burn; avoids burn → wave → burn churn; value from usage and rebalancing.
4. **dApp caps** — Incentives bounded; no runaway dApp payouts.
5. **Fair launch** — Specific initial supply; aligned long-term holders.
6. **Transparent parameters** — Floor, wave size, and fee split on-chain and verifiable.

Unlimited Supply Alternative: Bounded Inflation

An uncapped supply is often frowned upon, but it **can work** if the system is designed for predictability, decentralization, and value preservation. Some major L1s have no hard cap and remain sustainable via low issuance and fee burn. The key is **bounded inflation** and **on-chain guardrails**, not trust in any individual.

Principle	Implementation
Low, predictable inflation	Hard-coded emission schedule (e.g. 1% floor max). No governance can exceed bounds.
Fee burn	A portion of fees is burned. At high usage, burn can exceed emission → net deflation despite no supply cap.
Rules in stone	Emission bounds, burn rate, and governance limits are protocol parameters — transparent and auditable. No admin keys or emergency overrides.
Don't trust, verify	Community can verify all parameters on-chain. Trust is in code and decentralization, not in any person or team.
Decentralized governance	Parameter changes require broad consensus. No single party can alter inflation or supply dynamics unilaterally.

When unlimited supply makes sense: If the goal is long-term validator incentivization with minimal dilution risk, an uncapped supply with (1) very low inflation floor, (2) fee burn that can outpace emission, and (3) transparent, on-chain rules can be sustainable. The community can trust the system because it is verifiable, not because of promises.

Guardrails for unbounded supply (optional model):

- Inflation never exceeds X% per year (e.g. 1%) — hard-coded
- Fee burn rate is fixed or governance-bounded (e.g. 5–20%)
- No centralized minting; emission is algorithmic and auditable
- All parameters visible on-chain; dashboards and explorers show real-time supply and burn

Chosen Model: Uncapped Supply with Floor-Triggered Waves

> **Implementation note (audit #61 item 40):** shipped tokenomics enforce a **hard cap** of `MAX_SUPPLY` (1 billion BOING). Block emission is clamped so total native held (`balance + stake`) cannot exceed the cap. The floor emission formula remains nonzero on paper, but minting stops at the cap. The historical "uncapped / floor-triggered waves" design below is retained for context; mainnet parameters may still evolve via governance.

Boing's earlier design used an **uncapped supply** with a **floor-triggered rebalancing** mechanism:

Phase	Description
Genesis	Start with a **specific initial supply** (e.g. fixed amount at mainnet launch). No infinite mint at day one.
Ongoing	Normal activity: validators and treasury earn from fees. Circulating supply may decrease slowly (e.g. lost keys, optional burn if enabled) until the floor is reached.
Floor	When circulating supply reaches a **defined floor** (or a threshold near it), the protocol treats this as a rebalancing trigger.
New wave	A **new wave of tokens** is minted (size and destination defined by protocol/governance) to restore balance — e.g. validator incentives, treasury, or staking rewards — so the network remains secure and sustainable without arbitrary permanent deflation.

Design principles for this model:

- **Transparency:** Floor level, wave size, and trigger conditions are **on-chain and public**. No hidden minting.
- **Predictability:** Rules are clear: "When circulating supply ≤ X, mint wave Y according to Z." Community can verify and plan.
- **Decentralization:** Trigger and wave parameters are either **algorithmic** (code-defined) or **governance-set within bounds**. No single key can mint at will.
- **Balance:** The goal of each wave is to **create balance** — enough supply for staking rewards, ecosystem incentives, and security, without runaway inflation.

Parameters to define (and keep transparent):

- Initial supply at genesis
- Floor (or threshold) that triggers a new wave
- Wave size (fixed, formula-based, or governance-capped)
- Destination of minted tokens (e.g. staking rewards pool, treasury, both)
- Minimum interval between waves (if any) to avoid gaming or excessive minting

This model gives long-term sustainability without a hard cap, while keeping supply growth tied to observable, verifiable rules.

Burn in a Floor-Triggered Model: Optional

With floor-triggered waves, **burn is not strictly necessary**. The logic:

- **If you burn:** Supply falls faster → you hit the floor sooner → you mint a new wave. So burn mainly drives *how often* waves happen. Net effect over time can be “burn X, then mint Y” — i.e. more minting cycles.
- **If you don’t burn:** Supply falls only from other sinks (e.g. lost keys, one-off sinks). You hit the floor less often; waves are rarer and more event-driven. The rebalance mechanism still works; you just don’t add a built-in deflationary driver that then requires more waves.

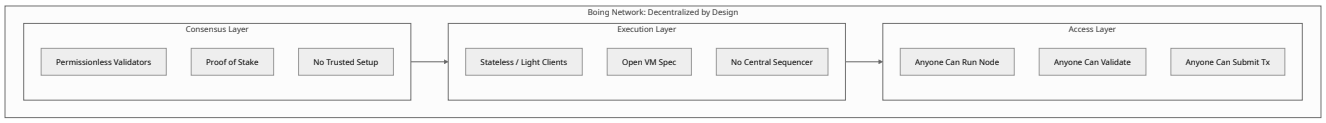
Conclusion: In this model, **burn is optional**. Skipping burn (or making it 0%) simplifies design and avoids the loop of “burn → floor → mint → burn again.” Fee revenue can go entirely to validators and treasury (e.g. 85% / 15% or 80% / 20%). Value comes from usage and sustainable waves, not from burn-induced scarcity. If you still want a small burn for narrative or between-wave deflation, it can be governance-adjustable and bounded (e.g. 0–5%).

Decentralization: A Non-Negotiable Requirement

Absolute decentralization must be a must when creating the Boing blockchain. This affects every layer of design:

Design Area	Decentralization Implications
Consensus	PoS over PoA; permissionless validator set; no trusted setup or central coordinator
Client Diversity	Support multiple independent implementations to avoid single-client dominance
Hardware Requirements	Keep validator and node hardware accessible (avoid mining/validator arms races)
State	Verkle trees enable stateless clients — lighter nodes = more participants
Governance	On-chain, transparent; avoid small-group control of parameters
Transparency	100% openness: protocol design, tokenomics, governance, treasury, security — all verifiable. Trust through verification.
Token Distribution	Fair launch; avoid concentrated initial allocation
Networking	Permissionless peer discovery; no whitelisted relayers or gatekeepers. Advanced: DHT + gossip-first, Sybil/eclipse resistance, incentivized relayers. See DECENTRALIZATION-AND-NETWORKING.md.

Decentralization Architecture Overview



Core Design Decisions

1. State Model — How Users & Contracts Store Data

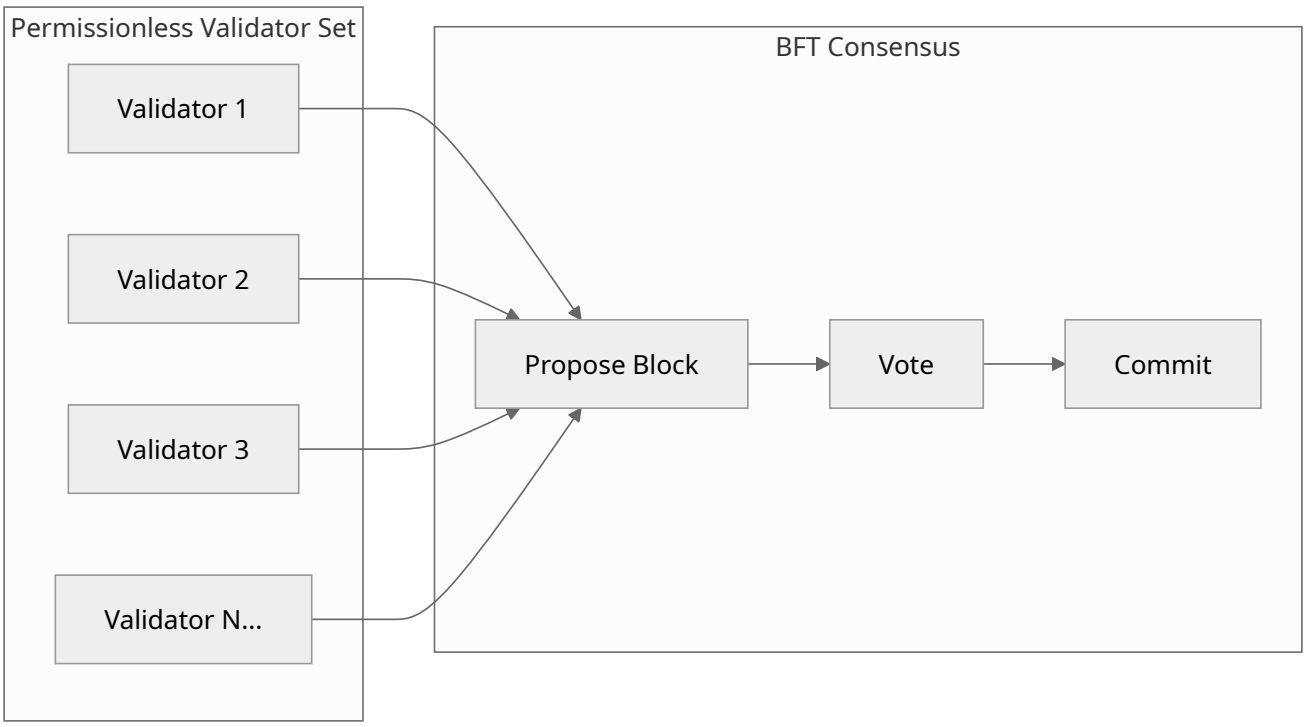
Model	Used By	Pros	Cons	Fit for Boing
Account model	Typical account-model L1s	Familiar, rich state, flexible	Harder to parallelize, state bloat	Good default for DeFi
UTXO model	Bitcoin, Cardano	Parallelizable, simpler proofs	Less expressive for smart contracts	Possible if you prioritize parallelism
Object-centric	Object-centric L1s (e.g. Sui)	Natural parallel execution	Different mental model, no stack-compatible bytecode	Strong if you want a novel UX

Recommendation: Account model with **explicit access lists** — familiar for DeFi while enabling parallelism without optimistic re-execution.

2. Consensus — How Validators Agree

Mechanism	Characteristics	Decentralization	Best For
Classic BFT (PBFT/Tendermint)	Leader-based, deterministic	High (permissionless validators)	Battle-tested, predictable finality
HotStuff / 2-chain BFT	Linear rounds, cleaner than PBFT	High	Modern BFT
Proof of History (PoH)	Verifiable time ordering	High	Throughput, parallel exec
Avalanche	Probabilistic metastability	High (any participant)	Fast finality, large validator set
Narwhal-Bullshark (DAG)	DAG mempool + ordering	High	Maximum throughput
Proof of Stake	Validator set by stake	High (permissionless entry)	Decentralization, economics

Recommendation: PoS + BFT — proof of stake for validator selection (permissionless, stake-based) and BFT for consensus (provably secure, deterministic finality).



3. Execution / Virtual Machine — Smart Contract Runtime

Approach	Pros	Cons
Custom VM (new bytecode)	Maximum control, tailor for your use cases	No ecosystem, tooling from scratch
WASM-based	Multi-language, standardized, deterministic	Must enforce determinism strictly
Move	Designed for safety, parallel execution	Diem/Meta lineage, licensing concerns
Custom IR + custom VM	Full design freedom	Most R&D intensive

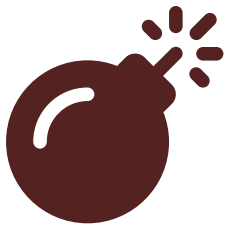
Recommendation: Custom deterministic VM — either **WASM-inspired** (reuse spec, define your own semantics) or **fully custom** for maximum authenticity.

Critical: Execution must be **deterministic** — same inputs always produce same outputs across all nodes.

4. Parallel Execution — Scaling Throughput

Strategy	How It Works	Used By
Declared dependencies	Tx declares which accounts/objects it touches; scheduler runs independent txs in parallel	High-throughput parallel L1s
Optimistic (Block-STM)	Execute in parallel, detect conflicts, re-execute conflicted txs	Aptos, Monad
Sequential	One tx at a time	Bitcoin, early account-model L1s

Recommendation: Declared dependencies — predictable performance, no re-execution overhead; fits well with account model + access lists.



5. State Storage — Data Structures for Efficiency

Structure	Pros	Cons
Merkle Patricia Trie	Battle-tested; common on legacy account L1s	Large proofs (~3KB), slow sync
Verkle trees	Tiny proofs (~200 bytes), enables stateless clients	Newer, complex crypto
Sparse Merkle	Simpler, compact	Less advanced than Verkle
Custom	Can optimize for your exact use case	R&D cost

Recommendation: Verkle trees — aligns with stateless-client research on major L1s; enables stateless validation and lighter nodes → **better decentralization**.

6. Block / Ledger Structure

Structure	Characteristics
Linear chain	Simple, familiar, easy to reason about
DAG	Higher throughput, more complex finality
Hybrid	DAG for ordering, chain for finality (Narwhal-Bullshark)

Recommendation: Start with **linear chain**; consider DAG-style batching later if needed.

Borrowing Optimal Ideas (Without Dependencies)

You can adopt **concepts** from other chains without depending on their codebases:

Domain	Idea	Source	Why It's Optimal
Consensus	HotStuff / 2-chain BFT	Research / Diem	Simpler than PBFT, provably correct
Ordering	Proof of History / VDF	Parallel L1 research	Enables parallel scheduling at scale
State	Verkle trees	Public L1 research	Tiny proofs, stateless clients
Parallelism	Declared dependencies	Parallel-oriented L1s	No rollbacks, predictable
Language design	Resource-oriented types	Move	Prevents double-spend bugs
Networking	libp2p	Many chains	Battle-tested P2P
Cryptography	Ed25519, BLS, secp256k1	Standards	Audited, no custom curves

Using an idea ≠ depending on a chain. Implement these yourself; the architecture remains Boing-native.

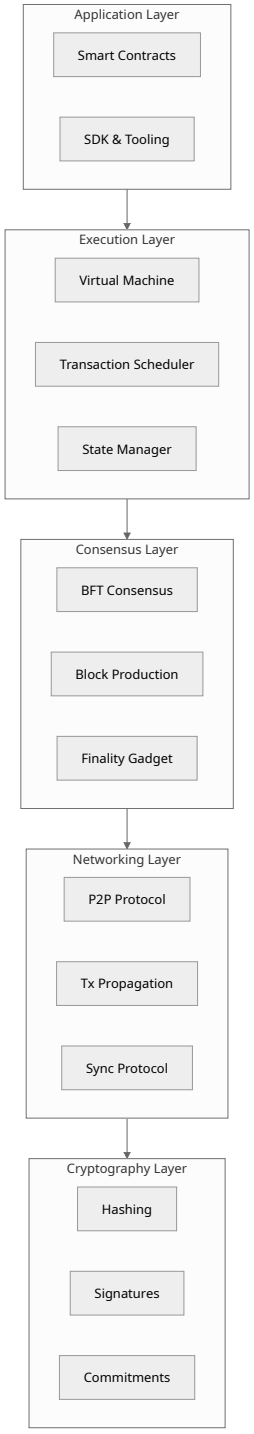
Tech Stack & Implementation

Language Choice

- **Rust** — Performance, memory safety, strong crypto ecosystem; industry standard for consensus/execution
- **Go** — Great for networking; some chains use it for node infrastructure

Recommendation: Rust for the core protocol (consensus, VM, state).

Full Stack Architecture



Component Responsibilities

Layer	Components	Purpose
Application	Smart contracts, SDK	Developer & user-facing logic
Execution	VM, scheduler, state manager	Run transactions, manage state
Consensus	BFT, block production, finality	Agreement among validators
Networking	P2P, propagation, sync	Node communication
Cryptography	Hashing, signatures, commitments	Security primitives

Phased Discovery & Build Plan

Phase 1: Design Lock (2–4 months)

1. **Design document** — State model, consensus, VM, parallelism, state structure
2. **Prototypes** — Consensus in a test harness; minimal VM; state tree ops
3. **Decisions** — Lock core choices and document rationale

Phase 2: Core Implementation (6–12 months)

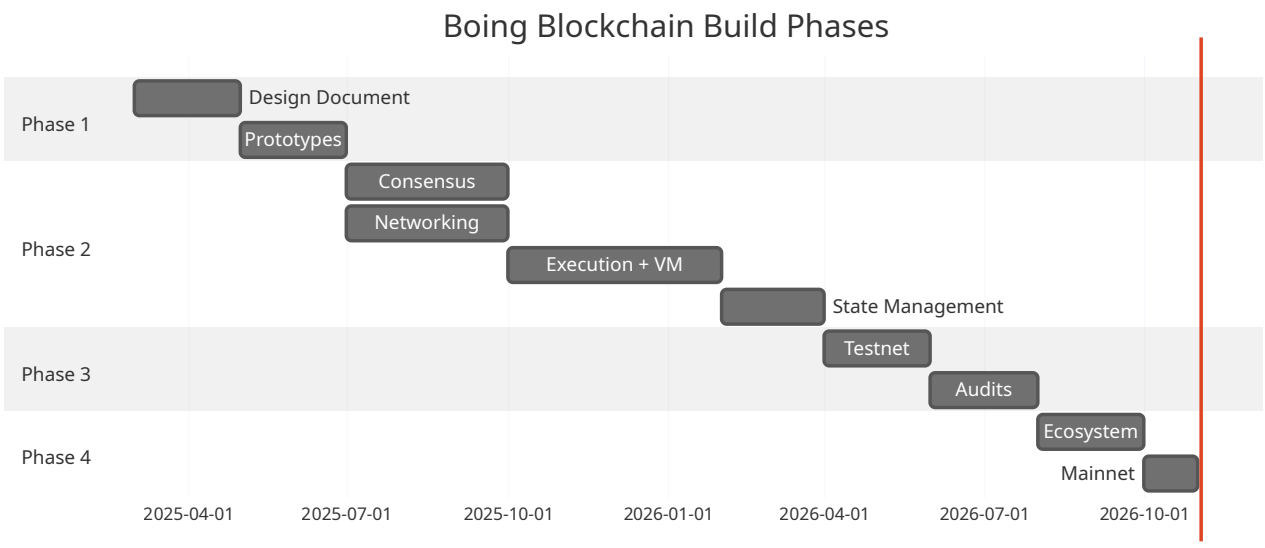
1. Consensus
2. Networking (P2P, sync)
3. Execution engine + VM
4. State management
5. End-to-end transaction flow

Phase 3: Integration & Testnet (3–6 months)

1. Multi-node testnet
2. External validator onboarding
3. Security audits
4. Performance and stress tests

Phase 4: Ecosystem & Mainnet

1. Developer docs, SDK, tooling
2. Wallets, explorers, RPC
3. Bridges (e.g., to boing.finance)
4. Mainnet launch and governance


Boing Blockchain Build Phases
Phase 1
Phase 2
Phase 3
Phase 4
Design Document
Prototypes
Consensus
Networking
Execution + VM
State Management
Testnet
Audits
Ecosystem
Mainnet
2025-04-01
2025-07-01
2025-10-01
2026-01-01
2026-04-01
2026-07-01
2026-10-01

Suggested "Authentic Boing" Configuration

A coherent configuration that prioritizes **authenticity**, **efficiency**, and **absolute decentralization**:

Component	Choice	Rationale
State model	Account model + explicit access lists	Familiar DeFi, enables parallelism
Consensus	PoS + HotStuff-style BFT	Decentralized, permissionless validators
Ordering	VDF or BFT-based ordering	Independent of PoH; still parallel-friendly
VM	Custom deterministic VM	Authentic; no foreign VM dependency
Parallelism	Declared dependencies (access lists)	Predictable, no re-execution
State tree	Verkle trees	Stateless clients, lighter nodes
Language	Rust	Performance, safety, tooling

Decentralization Checklist for Boing Config

- ☐ Validator set is permissionless (meets stake + technical requirements)
- ☐ No trusted setup or central key ceremony
- ☐ Stateless / light client support for broad participation
- ☐ Reasonable hardware requirements for validators and nodes
- ☐ No central sequencer or relay
- ☐ Open governance and parameter updates

Practical Considerations

Factor	Notes
Scope	Multi-year effort; consensus + VM + networking + crypto
Team	Requires distributed systems, crypto, and systems programming expertise
Security	Audits essential before mainnet; **continuous** independent audits (not one-time). See DEVELOPMENT-AND-ENHANCEMENTS.md.
Compatibility	Boing-native VM = full design freedom; plan for bridges if needed
DeFi Integration	Align with boing.finance as primary use case
Community	Actively foster validators, developers, users; clear documentation; educational programs; grant programs and hackathons. Make it easy to run nodes and contribute.

Discovery Checklist

Use this to drive design decisions:

Protocol

1. ☐ **Priorities** — Throughput vs. decentralization vs. simplicity vs. time-to-market
2. ☐ **State model** — Account vs. UTXO vs. object-centric
3. ☐ **Consensus** — Specific BFT variant and finality model
4. ☐ **VM** — Custom vs. WASM-inspired
5. ☐ **Parallelism** — Declared vs. optimistic
6. ☐ **Decentralization review** — Ensure each layer supports permissionless participation

UX & Human-Centered 7. [ ] **Account abstraction** — Native AA design; gasless, social recovery, session keys 8. [ ] **Trust & verification** — Human-readable intent; simulation; attestation 9. [ ] **MEV & cost** — Fair ordering; predictable gas; private mempool options

Technical Innovations 10. [ ] **Cross-chain primitives** — Light client verification; bridge standards 11. [ ] **Stateless clients** — Verkle proofs; compact verification 12. [ ] **Intent & simulation** — Pre-flight APIs; pre-confirmation design

Sustainability & Bootstrap 13. [ ] **Tokenomics** — Emission curve; fee split; no reward cliffs 14. [ ] **Design doc** — 1–2 page "Boing Design Principles" 15. [ ] **Prototype** — One critical component (consensus or VM) to validate approach

Deepening Decentralization & Enhancement Vision

Boing commits to pushing beyond baseline decentralization and amplifying authenticity:

- **DECENTRALIZATION-AND-NETWORKING.md** — Advanced P2P: DHT + gossip-first peer discovery, bootnode rotation, Sybil/eclipse resistance, WebRTC signaling for browser light clients, incentivized relayers, VDF/VRF for randomness, trustless bridges, network topology monitoring.
- **DEVELOPMENT-AND-ENHANCEMENTS.md** — Intent-based execution, dynamic fee allocation & developer royalties, decentralized storage (Filecoin/Arweave), Boing Studio IDE, AI-assisted SDK, continuous security audits, community engagement.
- **SECURITY-STANDARDS.md** — Layered security: protocol (consensus, crypto, VM, state), network (P2P, DDoS, Sybil/eclipse), application (SDK, automation), operational (audits, bug bounties, incident response).

Document Version: 1.5
Boing Network — Quality, enforced by the protocol.